

Task 21: Online Stock Span

Problem Description

Design an algorithm that collects daily price quotes for some stock and returns the span of that stock's price for the current day. The span of the stock's price in one day is the maximum number of consecutive days (starting from that day and going backward) for which the stock price was less than or equal to the price of that day.

Java Solution

```
class StockSpanner {
    Stack<int[]> stack;

    public StockSpanner() {
        stack = new Stack<>();
    }

    public int next(int price) {
        int span = 1;
        while (!stack.isEmpty() && stack.peek()[0] <= price) {
            span += stack.pop()[1];
        }
        stack.push(new int[]{price, span});
        return span;
    }
}
```